Bammidi Ketan Rao

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Education

National Institute of Technology Karnataka, Surathkal — Mangalore, Karnataka
Master of Technology – Computer Science and Engineering ***CGPA: 8.23*** — *Aug. 2025 – Present*

Visvesvaraya Technological University — Bangalore, Karnataka
Bachelor of Engineering – Computer Science and Engineering ***CGPA: 8.70*** — *Aug. 2019 – May 2023*

Experience

Software Engineer — Aug. 2023 – Aug. 2025
Tata Consultancy Services — *Bengaluru, India*

- Delivered **20+ enterprise features** by developing **REST APIs**, business workflows, and backend integrations, improving operational efficiency by **25%**.
- Engineered scalable backend services and **Kafka**-based event-driven workflows to support reliable, high-throughput enterprise application processing.
- Automated **CI/CD** pipelines using **Jenkins** and **Terraform**, improving deployment reliability and reducing production downtime through enhanced monitoring and release automation.
- Owned the end-to-end **Pega** training environment, ensuring platform stability, issue resolution, and seamless execution across enterprise training programs.

Projects

OmniForge | *Python, FastAPI, React, FAISS, Docker* — Jul. 2026 – Present

- Architected an **AI-powered brand intelligence platform** using **Retrieval-Augmented Generation (RAG)**, **FAISS**, and **local LLMs** for context-aware enterprise content generation.
- Developed scalable **FastAPI REST APIs** for brand management, campaign generation, real-time streaming, and feedback workflows.
- Built a responsive **React + TypeScript** dashboard with live AI responses, campaign history, and collaborative brand workspaces.
- Integrated **llama.cpp** for private **on-device LLM inference**, enabling low-latency AI processing without external cloud AI services.

ATLASS | *Python, FastAPI, React, FAISS, PostgreSQL, HuggingFace, PyTorch* — Sep. 2025 – Dec. 2025

- Built an **AI-powered research platform** for scientific paper understanding, **semantic search**, and knowledge discovery using **LLM-assisted retrieval**.
- Engineered a **hybrid retrieval pipeline** combining **FAISS vector search**, **BM25 reranking**, and metadata-aware indexing for high-relevance document retrieval.
- Automated generation of **system specifications**, implementation blueprints, and **PyTorch baselines** from retrieved research literature to accelerate AI experimentation.
- Designed and developed a scalable **FastAPI REST API** backend with a **React** frontend, integrating **HuggingFace Transformers**, PostgreSQL, and asynchronous request processing.

HyperQ – Distributed Queue | *C++17, Multithreading, Linux, Distributed Systems* — Feb. 2025 – Present

- Built a **Kafka-inspired distributed message queue** in **C++17** featuring a broker-based **publish-subscribe architecture** for scalable and reliable message processing.
- Implemented **partitioned append-only commit logs**, **consumer groups**, and **offset management** to ensure durable storage, ordered delivery, message replay, and fault recovery.
- Engineered **thread-safe concurrent workflows** using **mutexes**, **condition variables**, and modern C++ concurrency primitives; designed a **replication-ready distributed architecture** for horizontal scalability.

Technical Skills

Languages: C++, C, Python, Java, SQL
Frameworks & Libraries: FastAPI, React, TypeScript, PyTorch, HuggingFace Transformers, FAISS
Systems & Tools: Linux, Docker, Git, CMake, PostgreSQL, REST APIs, llama.cpp
Core Concepts: Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Computer Networks, Multithreading, Concurrency, Synchronization, Distributed Systems, Memory Management